In [1]:
```
import whisper
import pandas as pd
import os
```

In [17]:
```
data_folder=r'..\data\audiodata'
all_dfs=[]
for file in  os.listdir(data_folder):
    if file.endswith('.csv'):
        path=os.path.join(data_folder,file)
        df=pd.read_csv(path,encoding='utf-8')
        df=df.iloc[0:30,:3]
        df['participant']=file #mark which participant(file)
        all_dfs.append(df)
data=pd.concat(all_dfs,ignore_index=True)
data.head()
```

Out[17]:

	Sentence	Stimulus	Transcription	participant
0	1.0	Quiero cortarme el pelo (7)	NaN	participant1.csv
1	2.0	El libro está en la mesa (7)	NaN	participant1.csv
2	3.0	El carro lo tiene Pedro (8)	NaN	participant1.csv
3	4.0	El se ducha cada mañana (9)	NaN	participant1.csv
4	5.0	¿Qué dice usted que va a hacer hoy? (9)	NaN	participant1.csv

In [18]:
```
data.shape
```

Out[18]: (120, 4)

In [19]:
```
import re
```

In [20]:
```
def clean(text):
    return re.sub(r"\s*\(.*?\)","",str(text)).strip()
```

In [21]:
```
data['Stimulus']=data['Stimulus'].apply(clean)
data.head()
```

Out[21]:

	Sentence	Stimulus	Transcription	participant
0	1.0	Quiero cortarme el pelo	NaN	participant1.csv
1	2.0	El libro está en la mesa	NaN	participant1.csv
2	3.0	El carro lo tiene Pedro	NaN	participant1.csv
3	4.0	El se ducha cada mañana	NaN	participant1.csv
4	5.0	¿Qué dice usted que va a hacer hoy?	NaN	participant1.csv

In [42]:
```
model=whisper.load_model('small')
```

In [53]:
```
import os
folder = r'..\data\audio'
transcriptions = []
```

```
for file in os.listdir(folder):
    if file.endswith(".mp3"):
        audio_path = os.path.join(folder, file)
        print("processing",file)
        result = model.transcribe(audio_path, language="es")
        full_text = result["text"]
        sentences = re.split(r'[.!?]', full_text)
        sentences = [s.strip() for s in sentences if s.strip()]
        transcriptions.extend(sentences)
len(transcriptions)
```

```
processing participant1.mp3
processing participant2.mp3
processing participant3.mp3
processing participant4.mp3
```

Out[53]: 116

In [56]:
```
data["Transcription"].iloc[:len(transcriptions)] = transcriptions
data.head()
```

```
C:\Users\USER\AppData\Local\Temp\ipykernel_6804\1683590571.py:1: SettingWithCopyW
arning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy
  data["Transcription"].iloc[:len(transcriptions)] = transcriptions
```

Out[56]:

	Sentence	Stimulus	Transcription	participant
0	1.0	Quiero cortarme el pelo	Quiero cortarme el pelo	participant1.csv
1	2.0	El libro está en la mesa	Quiero cortarme el pelo	participant1.csv
2	3.0	El carro lo tiene Pedro	El libro está en la mesa	participant1.csv
3	4.0	El se ducha cada mañana	El libro está en la mesa	participant1.csv
4	5.0	¿Qué dice usted que va a hacer hoy?	El caro lo tiene el pelo	participant1.csv

Note on Audio Segmentation

During transcription, the Whisper model produced 116 segments instead of the expected 120 sentences. This occurred because the audio recordings contain long pauses and some irrelevant audio at the beginning, which affect how Whisper automatically segments speech. Since the model splits audio based on pauses rather than the experimental sentence boundaries, the resulting number of segments may differ from the expected number of stimulus sentences. To address this, the audio was trimmed to remove unnecessary starting audio, but some segmentation inconsistencies remained.

In [57]:
```
data.to_csv("../results/participant_transcription.csv", index=False)
```

APPROACH

The goal of Task I is to generate accurate transcriptions of learner responses from Spanish elicited imitation task recordings. For this purpose, I used the OpenAI Whisper automatic speech recognition (ASR) model, a multilingual transformer-based model that performs well for Spanish speech recognition.

Each participant audio recording was processed using Whisper to produce a text transcription. The model was configured with the Spanish language parameter to improve recognition accuracy.

Whisper automatically segments audio based on pauses in speech. However, these segments do not always correspond exactly to the 30 expected stimulus sentences in the dataset. Therefore, the full transcription of the recording was generated and then aligned with the rows in the provided spreadsheet.

The transcription results were inserted into the Transcription column of the dataset while preserving the original learner production. According to the task instructions, grammatical or lexical errors produced by participants were not corrected; only transcription errors from the ASR system were adjusted if necessary.

The final output is an Excel file containing the following columns:

| Sentence | Stimulus | Transcription |

Evaluation of Transcription Output

The transcription results were evaluated through manual inspection. Each sentence was reviewed to ensure that the transcription accurately reflects the participant's spoken response.

The evaluation focused on:

- correctness of recognized words
- preservation of learner errors and disfluencies
- alignment with the expected number of sentences in the dataset

Minor orthographic variations, such as missing accent marks in Spanish words, were considered acceptable when they did not affect the meaning of the transcription.